FINITENESS OF THE COHOMOLOGY OF THE EXTENSION BY ZERO OF A LOCALLY CONSTANT SHEAF ANNIHILATED BY A PRIME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the prime-to-l finite etale cover and the trace-method splitting step, the Zariski-factorization step, the shriek/pushforward identity used to move the extension by zero across the finite morphism, and the filtration-plus-short-exact-sequence reduction to the constant case.

Editorial note. The label lemma-vanishing-easier-coefficients names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source etale-cohomology.tex –label lemma-vanishing-easier-coefficients –start 16010 –end 16059 –local-id 3553 returned RESULT=OK (L3553 free, occupancy FREE, public interval clear). Candidate taken from lane-s4/halfB-priority.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The trace method and the finite etale cover of degree prime to l. Same chapter, the section on the trace method and the lemma producing a finite etale V -> U of degree prime to l together with a filtration of the pullback. Supplies the cover f : V -> U of degree prime to l and the maps G -> f_*f^{-1}G -> G whose composition is an isomorphism, and the resulting filtration of the pulled-back coefficient sheaf.
- Zariski's main theorem, quasi-finite separated version. More on Morphisms, the lemma that a quasi-finite separated morphism factors as an open immersion followed by a finite morphism. Supplies the diagram with Y -> X finite and j' : V -> Y an open immersion with dense image.
- Shriek pushforward versus pushforward for a finite morphism. Same chapter, the lemma identifying j_! composed with the pushforward along a finite morphism with the pushforward of the shriek extension. Supplies the identity j_!f_*f^{-1}G = fbar_*j'_!f^{-1}G used to replace the extension by zero along j by an extension by zero along j'.
- Pushforward of finite type coefficient sheaves along a finite morphism. Same chapter, the lemma reducing finiteness of the cohomology of a pushforward along a finite morphism to the source. Supplies the reduction from

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j'_!f^{-1}G on Y to the sheaf j'_!f^{-1}G on Y with the same finiteness conclusions.
- Finiteness for a finite constant coefficient sheaf, and the short exact sequence reduction. Same chapter, the lemma treating the case of a constant finite coefficient sheaf and the lemma reducing along a short exact sequence of coefficient sheaves. Supplies the final reduction from a filtered coefficient sheaf with finite quotients to the constant case M.

Source reference. Original reference 0GJ1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11612–11612. Prerequisite pointer invoked at proof line(s) 16047; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GJD: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15909–15909. Prerequisite pointer invoked at proof line(s) 16051; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GJ0: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10479–10479. Prerequisite pointer invoked at proof line(s) 16028; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GJG: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15997–15997. Prerequisite pointer invoked at proof line(s) 16055; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GJF: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15973–15973. Prerequisite pointer invoked at proof line(s) 16058; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05K0: sources/stacks-chow/tags.txt, source lines 7184–7184. Prerequisite pointer invoked at proof line(s) 16037; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 03SH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10330–10330. Prerequisite pointer invoked at proof line(s) 16029; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let Λ be a Noetherian ring, let k be an algebraically closed field, let X be a separated, finite type scheme of dimension ≤ 1 over k, let $j : U \to X$ be an open immersion with U connected, let ℓ be a prime number, let $n > 0$, and let $\mathcal{G}$ be a finite type, locally constant sheaf of Λ-modules on $U_{étale}$ annihilated by ℓ^n. Then*

(1) *$H^q_{étale}(X, j_!\mathcal{G})$ is a finite Λ-module if ℓ is prime to $char(k)$,*
(2) *$H^q_{étale}(X, j_!\mathcal{G})$ is a finite Λ-module if X is proper.*

Proof. Let $f : V \to U$ be a finite étale morphism of degree prime to ℓ as in Lemma 0GJ0. The discussion in Section 03SH gives maps

$$\mathcal{G} \to f_* f^{-1}\mathcal{G} \to \mathcal{G}$$

whose composition is an isomorphism. Hence it suffices to prove the finiteness of $H^q_{étale}(X, j_! f_* f^{-1}\mathcal{G})$. By Zariski's Main theorem (More on Morphisms, Lemma 05K0) we can choose a diagram

$$\begin{array}{ccc} V & \xrightarrow{j'} & Y \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \overline{f}} \\ U & \xrightarrow{j} & X \end{array}$$

with $\overline{f} : Y \to X$ finite and j' an open immersion with dense image. Since f is finite and V dense in Y we have $V = U \times_X Y$. By Lemma 0GJ1 we have

$$j_! f_* f^{-1}\mathcal{G} = \overline{f}_* j'_! f^{-1}\mathcal{G}$$

By Lemma 0GJD it suffices to consider $j'_! f^{-1}\mathcal{G}$. The existence of the filtration given by Lemma 0GJ0, the fact that $j'_!$ is exact, and Lemma 0GJG reduces us to the case $\mathcal{F} = j'_! \underline{M}$ for a finite Λ-module M which is Lemma 0GJF. □